

A finite-dimensional nonclosed zero set for a sum of maximally monotone operators

Candidate full solution packet for arXiv:1110.4877

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Status. Candidate full solution, likely valid. The construction gives an affirmative answer to Remark 3.12 of Bauschke–Boţ–Hare–Moursi.

1 Source question

Bauschke, Boţ, Hare, and Moursi study the primal solution set $Z = \text{zer}(A + B)$ for maximally monotone operators on a real Hilbert space [1]. Their infinite-dimensional example shows that Z need not be closed. Remark 3.12 asks whether this can already happen in finite dimension when $\text{dom } A \cap \text{dom } B \neq \emptyset$.

Transcription. “We do not know whether or not such counterexamples can reside in finite-dimensional Hilbert spaces when $\text{dom } A \cap \text{dom } B \neq \emptyset$.” The source further observes that at least one operator must be nonparamonotone and that standard relative-interior sum qualifications must fail.

proof is very elegant and completely different from the above Passty-like proof. In their 2009 follow-up paper [23], Eckstein and Svaiter generalize the notion of the extended solution set to three or more operators; their corresponding proof of convexity in [23, Proposition 2.2] is more direct and along Passty's lines.

Remark 3.10 (convexity of Z and K) If Z is nonempty and a constraint qualification holds, then $A + B$ is maximally monotone (see, e.g., [5, Section 24.1]) and therefore $Z = \text{zer}(A + B)$ is convex. It is somewhat surprising that Z is always convex even without the maximal monotonicity of $A + B$.

One may inquire whether or not Z is also closed, which is another standard property of zeros of maximally monotone operators. The next example illustrates that Z may fail to be closed.

Example 3.11 (Z need not be closed!) Suppose that $X = \ell_2$, the real Hilbert space of square-summable sequences. In [10, Example 3.17], the authors provide a monotone discontinuous linear at most single-valued operator S on X such that S is maximally monotone and its adjoint S^* is a maximally monotone single-valued extension of $-S$. Hence $\text{dom} S$ is not closed. Now assume that $A = S$ and $B = S^*$. Then $A + B$ is operator that is zero on the dense proper subspace $Z = \text{dom}(A + B) = \text{dom} S$ of X . Thus Z fails to be closed. Furthermore, in the language of Passty's parallel sums (see Remark 3.8), this also illustrates that the parallel sum need not map a point to a closed set.

Remark 3.12 We do not know whether or not such counterexamples can reside in finite-dimensional Hilbert spaces when $\text{dom} A \cap \text{dom} B \neq \emptyset$. On the one hand, in view of the forthcoming Corollary 5.3(i), any counterexample must feature at least one operator that is not paramonotone, which means that the operators cannot be simultaneously subdifferential operators of functions in Γ . On the other hand, one has to avoid the situation when $A + B$ is maximally monotone, which happens when $\text{ri} \text{dom} A \cap \text{ri} \text{dom} B \neq \emptyset$. This means that neither is one of the operators allowed to have full domain, nor can they simultaneously have relatively open domains, which excludes the situation when both operators are maximally monotone linear relations (i.e., maximally monotone operators with graphs that are linear subspaces, see [9]).

Remark 3.13 We note that K and Z are in general not maximally monotone. Indeed if Z , say, is maximally monotone, then Corollary 3.7 and [9, Theorem 4.2] imply that $\text{gr} Z$ is actually affine (i.e., a translate of a subspace) and so are Z and K (as range and domain of Z). However, the set Z of Example 2.7 need not be an affine subspace (e.g., when U , V and Z coincide with the closed unit ball in X).

Figure 1: Source-PDF page 10: the infinite-dimensional example and the finite-dimensional question in Remark 3.12.

2 Definitions

For a set-valued operator $T : \mathbb{R}^2 \rightrightarrows \mathbb{R}^2$, write $\text{gra } T = \{(z, z^*) : z^* \in Tz\}$. It is *monotone* when

$$\langle z - w, z^* - w^* \rangle \geq 0 \quad ((z, z^*), (w, w^*) \in \text{gra } T),$$

and *maximally monotone* when its graph has no proper monotone extension. We use Minty's criterion: a monotone T on a Hilbert space is maximally monotone if and only if $\text{ran}(\text{Id} + T)$ is the whole space [2].

Let $H = \mathbb{R} \times (-\infty, 0]$ and $L = \mathbb{R} \times \{0\}$. Their normal cones use the outward-normal convention; in particular,

$$N_H(x, y) = \begin{cases} \{(0, 0)\}, & y < 0, \\ \{0\} \times [0, \infty), & y = 0, \end{cases} \quad N_L(x, 0) = \{0\} \times \mathbb{R}.$$

3 Proof intuition

To make the zero set an open ray, let one operator live on the lower half-plane but force its first coordinate to stay positive; let the second operator be the normal cone to the horizontal axis. The only difficult point is maximality of an operator with nonclosed domain. The vector field

$$F(x, y) = \left(\frac{y}{x^2}, \frac{1}{x} \right)$$

is designed for this: its off-diagonal derivatives are skew and cancel in the symmetric Jacobian, while the singularity at $x = 0$ prevents any missing graph point. More decisively, the resolvent equation reduces to a scalar continuous function whose endpoint limits cover every target in $\mathbb{R}^2$.

4 Counterexample

Theorem 1. *There exist maximally monotone operators $A, B : \mathbb{R}^2 \rightrightarrows \mathbb{R}^2$ with $\text{dom } A \cap \text{dom } B \neq \emptyset$ such that*

$$\text{zer}(A + B) = (0, \infty) \times \{0\}.$$

Consequently, the primal solution set of a sum of two maximally monotone operators need not be closed even in a finite-dimensional Hilbert space.

Proof. Set $D = (0, \infty) \times (-\infty, 0]$. For $(x, y) \in D$, put

$$F(x, y) = \left(\frac{y}{x^2}, \frac{1}{x} \right),$$

and define

$$A(x, y) = F(x, y) + N_H(x, y) \quad ((x, y) \in D), \quad A(x, y) = \emptyset \quad ((x, y) \notin D).$$

Thus, explicitly,

$$A(x, y) = \begin{cases} \{(y/x^2, 1/x)\}, & x > 0, y < 0, \\ \{0\} \times [1/x, \infty), & x > 0, y = 0, \\ \emptyset, & \text{otherwise.} \end{cases}$$

We first prove that A is maximally monotone. On the open half-plane $\{x > 0\}$,

$$DF(x, y) = \begin{pmatrix} -2y/x^3 & 1/x^2 \\ -1/x^2 & 0 \end{pmatrix}, \quad \frac{DF + DF^\top}{2} = \begin{pmatrix} -2y/x^3 & 0 \\ 0 & 0 \end{pmatrix}.$$

The latter matrix is positive semidefinite on D . Integrating along the line segment between any two points of the convex set D shows that F is monotone on D . The boundary term is a value of the monotone normal cone N_H , so A is monotone.

It remains to prove $\text{ran}(\text{Id} + A) = \mathbb{R}^2$. Fix an arbitrary $(p, q) \in \mathbb{R}^2$. If $q > 0$ and $p \geq 1/q$, take

$$(x, y) = (p, 0), \quad r = q - \frac{1}{p} \geq 0.$$

Then $(0, 1/p + r) \in A(p, 0)$ and $(p, q) = (p, 0) + (0, 1/p + r)$.

In all other cases seek an interior point by setting

$$y = q - \frac{1}{x} < 0.$$

The first coordinate of $(x, y) + F(x, y)$ equals

$$h_q(x) := x + \frac{q}{x^2} - \frac{1}{x^3}.$$

If $q > 0$ and $p < 1/q$, use $0 < x < 1/q$. Then

$$\lim_{x \downarrow 0} h_q(x) = -\infty, \quad \lim_{x \uparrow 1/q} h_q(x) = \frac{1}{q} > p.$$

Continuity gives $x \in (0, 1/q)$ with $h_q(x) = p$, and the definition gives $y < 0$. If $q \leq 0$, use $x \in (0, \infty)$; now

$$\lim_{x \downarrow 0} h_q(x) = -\infty, \quad \lim_{x \rightarrow \infty} h_q(x) = +\infty,$$

so again $h_q(x) = p$ for some $x > 0$, while automatically $y = q - 1/x < 0$. In either interior case,

$$(x, y) + F(x, y) = (h_q(x), q) = (p, q).$$

Hence $\text{ran}(\text{Id} + A) = \mathbb{R}^2$, and Minty's criterion proves that A is maximally monotone.

Now set $B = N_L$, which is maximally monotone. The common domain is

$$\text{dom } A \cap \text{dom } B = (0, \infty) \times \{0\}.$$

For $t > 0$,

$$A(t, 0) = \{0\} \times [1/t, \infty), \quad B(t, 0) = \{0\} \times \mathbb{R},$$

and therefore $(A + B)(t, 0) = \{0\} \times \mathbb{R}$ contains $(0, 0)$. Outside the common domain, $A + B$ is empty. It follows that

$$\text{zer}(A + B) = (0, \infty) \times \{0\},$$

which is not closed in $\mathbb{R}^2$. □

Consistency with the source obstruction. The operator A is not paramonotone. Indeed, fix $x > 0$ and distinct $y_1, y_2 < 0$. The monotonicity pairing between $((x, y_i), F(x, y_i))$ is zero because the displacement is vertical while the value difference is horizontal. But $F(x, y_2) \notin A(x, y_1)$, so the cross-membership required by paramonotonicity fails.

5 Verification and novelty boundary

The accompanying script samples graph pairs and arbitrary targets for the Minty range equation; it is only a stress check, since the proof above is exact. On 11 August 2026, the run indexes and bounded current-literature searches were queried using arXiv:1110.4877, the exact Remark 3.12 wording, “ $\text{zer}(A + B)$ not closed”, “maximally monotone”, and “finite-dimensional”. Nearby Attouch–Théra papers from 2013, 2016, and 2021 were checked at title/abstract level. No later exact answer or occurrence of this construction was found. This supports, but cannot guarantee, novelty.

Human review. The key check is the three-regime surjectivity argument for $\text{Id} + A$, including the endpoint $x = 1/q$ when $q > 0$. The normal-cone sign at $y = 0$ should then be verified. No computational or conditional dependency remains.

References

- [1] H. H. Bauschke, R. I. Boţ, W. L. Hare, and W. M. Moursi, *Attouch–Théra duality revisited: paramonotonicity and operator splitting*, arXiv:1110.4877 (2011), especially Remark 3.12.
- [2] G. J. Minty, *Monotone (nonlinear) operators in Hilbert space*, Duke Math. J. **29** (1962), 341–346.